Correctness Boundaries of Typed Plans for Natural-Language Robot Fleet Control

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Abstract

A fleet can accept a well-formed plan and still assign an extra robot, reject a requested route, or report success for only part of the original command. We examine these outcomes in a three-level Robot Operating System 2 implementation with twenty simulated robots. An offline benchmark of 44 cases checks plan structure and selected case-specific requirements. Across five local configurations without a response schema, 70.0–87.3% of responses pass. All runs in the selected 18-run simulation series meet their pose-based task conditions. That series includes two successful repeats that replaced two incomplete runs after a software change. The available plans and logs document an extra pickup assignment, a route rejection after the system assembled a line abreast in response to a column request, and an arrival report that omitted a requested robot. For carry tasks, the evaluator records pickup and drop-off zone presence for the same robot. The cases show why the original command, the action sent to the robots, and the observed outcome need to be tracked separately.

Keywords: Multi-robot systems, Language models, Fleet coordination, Typed plans, Deterministic guards, ROS 2

1 Introduction

In three recorded plans, the plan sent an extra robot to a pickup already assigned to another carrier. The plan format admitted both assignments (Fig. 1). Language interfaces can express actions in terms of skills and programs [1, 2]. For a fleet, the resulting assignments and goals still need to be coordinated across robots sharing the same workspace.

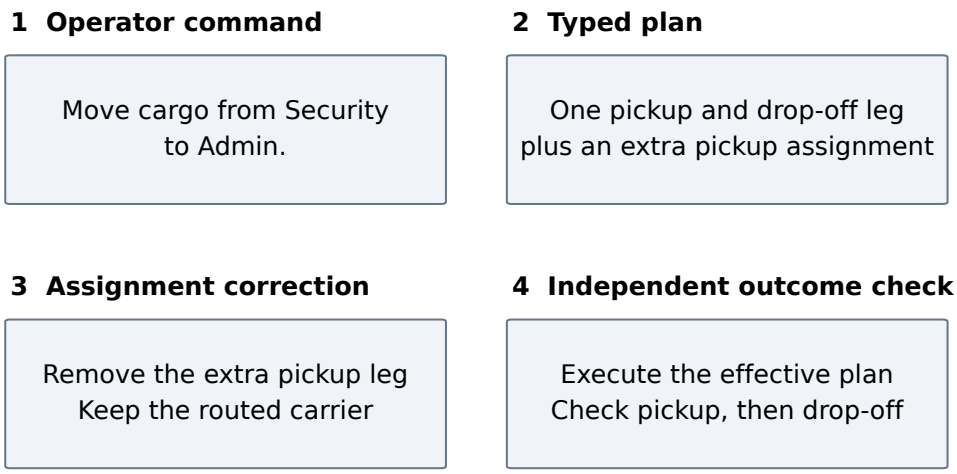


Fig. 1 Illustrative duplicate-carrier correction at the language–coordination boundary. An additional robot is routed to the same pickup. The guard retains the robot with the drop-off leg and removes the extra pickup assignment

We separate task interpretation from multi-agent path finding (MAPF). Direct language-model generation of MAPF solutions has failed on tested room and maze instances [3]. Our system instead sends fleet-level actions to a dedicated planner.

Hierarchical control, formal intermediate representations, and deterministic validation are established approaches. Our contribution is a case study of their coverage in a mobile fleet. We describe the checks at the interfaces and document execution cases in which a structurally accepted plan still exhibits an assignment error, a route rejection, or a mismatch between a component report and the original request.

We ask which validation criteria generated plans fail to satisfy in the tested configurations (RQ1), which assignment and spatial conflicts the implemented checks handle in the recorded cases (RQ2), and how a component can report success while part of the original requested action remains incomplete (RQ3). The offline benchmark addresses RQ1. The simulation plans, logs and task-state summaries provide evidence for RQ2 and RQ3.

2 Related Work

An instruction to a robot fleet leaves two different problems to solve: deciding what the robots should do and finding motions that realize those decisions. Classical multi-agent path finding addresses the latter once starts, goals and movement rules have been specified. Conflict-Based Search [4] resolves path conflicts through constraints, while Priority-Based Search (PBS) [5] and large-neighborhood repair (MAPF-LNS2) [6] offer alternative ways to coordinate paths. The resulting paths coordinate motion under the specified rules, but do not determine whether the supplied assignments and goals express a user's request. PlanBench documents failures of language models on its own plan-generation tasks [7].

Grounding language in executable skills is therefore central to early robot planners based on large language models (LLMs). SayCan [1] combines a language model's estimate of a skill's usefulness with an affordance estimate of whether the robot can execute it in the current state. This connects task interpretation to available behavior without asking the language model to generate motor commands. Code as Policies [2] gives the model a more expressive interface: it composes perception and control APIs into programs, including control flow and numerical operations. For fleet control the distinction matters: choosing an available action and assembling a coherent mission require different checks.

Execution can depart from a plan even when its actions are available. Inner Monologue [8] feeds observations and success reports back into language reasoning, allowing subsequent decisions to account for execution. ProgPrompt [9] incorporates this concern into program-like plans through assertions and recovery steps. In its simulated evaluation, assertions are assessed by an LLM using environment-state information, whereas our runtime guards apply explicit predicates. In both systems, feedback informs the next planning decision.

For a robot team, grounding individual actions must be combined with allocation and coordination [10]. The task-allocation taxonomy distinguishes single- and multi-task robots, single- and multi-robot tasks, and instantaneous and time-extended assignment [11]. Our duplicate-carrier case concerns the assignment of a single-robot task. SMART-LLM [12] uses prompted task decomposition, coalition formation and allocation to construct a team plan. This makes robot selection part of language-level planning rather than an input assumed to be correct. RoCo [13] instead uses dialogue between robot-specific language agents to develop a joint plan, which is checked before a centralized multi-arm planner generates motion. These designs place coordination at different points in the pipeline, but both need a connection between the proposed team behavior and its executor's capabilities; our duplicate-carrier and shared-destination cases examine that connection.

LaMMA-P and HiP-CL construct explicit planning problems for robot teams [14, 15]. LLM+P translates instructions into the Planning Domain Definition Language (PDDL) and delegates search to a classical planner [16]. LLM-Modulo uses external checks to evaluate model proposals [17]. Temporal-logic synthesis can give guarantees relative to a realizable specification and an environment model [18].

CommandSwarm checks generated behavior trees against allowed swarm primitives [19]. Sangchai et al. [20] provide an intermediate plan for human review. Khatana

et al. [21] ground candidate plans into trajectories and check for inter-robot conflicts. Our case study traces what changes between the original command and the action sent to the fleet, and what the resulting status reports establish.

3 System Overview

Figure 2 shows the operator-chat path. Level 1 turns a natural language command into a typed plan and applies interface checks. Level 2 coordinates the resulting fleet actions. Level 3 tracks the scheduled waypoints and executes the control commands.

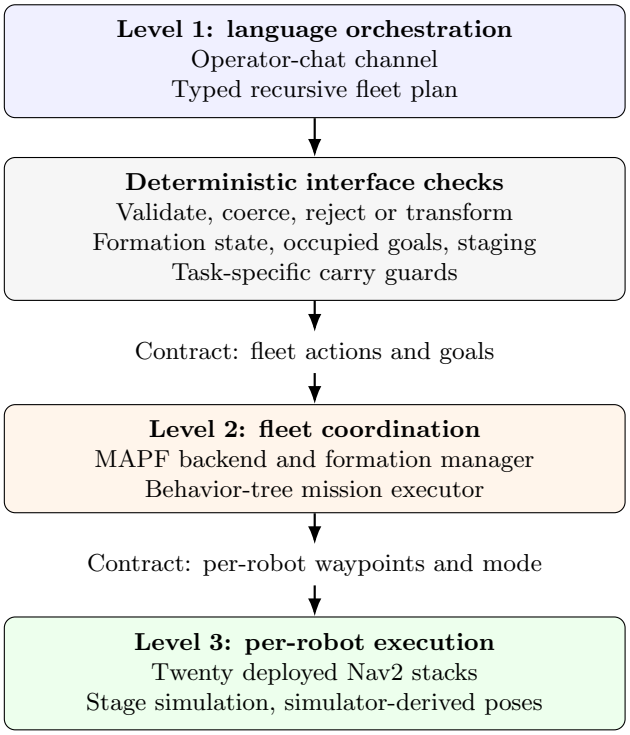


Fig. 2 Operator-chat path from a typed fleet plan through interface checks, fleet coordination, and per-robot execution

The interface exposes four distinct correctness questions. Structural validity concerns admissible node types and fields. Task meaning concerns the assignments and goals implied by the operator's command. Geometric feasibility concerns whether robots or formations can traverse the required space. Evaluator completion concerns whether the specified pose-based events occur. No implication between these properties follows from the typed representation alone. Table 1 links each check to the corresponding evidence.

Table 1 Implemented checks and their observed scope

Property	Check in this implementation	Evidence and limit
Plan representation	Recursive node and field checks, with selected type coercions	E1 also checks case-specific requirements. Acceptance leaves other mission properties open.
Robot assignments	Active-formation membership and single-carrier rules	Single-carrier rule fired in two selected M2 runs.
Occupied goals	Chat-level goal adjustment and planner-level blocked-cell substitution	Both mechanisms appear in M4 run `20260818_140918`. No chat rewrites in the 13 selected language runs.
Shared drop-off	Spread coincident goals for distinct carry tasks	Logged in three M3 runs. The design does not isolate its effect.
Formation assembly	Stage followers and check pose tolerances	M4 includes an assembled formation and a separate staging failure.
Formation route	Static path check with the planner's footprint approximation	LNS2 rejected the leader's route in M4 run `20260819_073009`. No pre-assembly check of the full oriented formation route.
Task completion	Pose samples in specified zones at 5 Hz, with pickup before drop-off	All 18 selected M2/M3 runs satisfy this evaluator. Physical transport and intent are outside the metric.
Repair and supervision	Execution remediation, verification repair and bounded mission supervision	One supervision pass per selected language run. No remediation or verification-repair attempts in that series.

Simulation environment

Twenty differential-drive robots operate in Stage [22] on the `amongus` map (66.6 × 37.75 m, Fig. 3). The world file places five groups of four around their respective home locations. The number deployed is distinct from the number moving in any interval.

4 Level 1: Language Orchestration

Plan grammar

In the operator-chat channel, a response contains a natural-language reply and a plan tree in a JavaScript Object Notation (JSON) object. Four leaf types cover fleet actions — `mapf` (move a robot set to a goal set), `formation` (assemble and hold), `disband`, and `idle` (returns no action and can carry an escalation reason). Two containers, `sequence` and `parallel`, nest recursively. This example illustrates the plan format:

```
{"type":"sequence","steps":[
 {"type":"mapf","robot_ids":[8,9,10,11],
  "goals":[[-8.7,-5.9],[-9.6,-5.9],
           [-8.7,-6.8],[-9.6,-6.8]]},
 {"type":"formation","formation_id":"column_8",
  "leader_ns":"robot_8",
  "follower_ns":["robot_9",
   "robot_10","robot_11"],
  "offsets_x":[-1.5,-3.0,-4.5],
  "offsets_y":[0,0,0]}]}
```


The finite set of node types supports recursive validation and a JSON schema. These checks cover the implemented structural rules, not arbitrary task semantics or geometric feasibility.

Three channels and grounding

The reactive channel returns a wait, abort, or replan decision to a behavior-tree node on warning or error. The proactive observer can send a flat mode command to the behavior tree. The operator-chat channel uses the recursive plan grammar and the guards described here. E1 exercises offline generation for that grammar, and the language runs in E2 use the operator-chat path. That path can query live pose and task state and compute room placements through eight allowlisted tools.

Generation and runtime checks

E1 compares text generation with and without a response schema. Both modes parse the returned plan and apply case validators. The E2 operator chat instead enables a tool-calling loop. Enabling that loop allows an auxiliary tool call but does not require one. In both settings the plan is returned as JSON.

The operator-chat path validates node types and fields, coerces selected values, guards active-formation membership, adjusts occupied goals, and stages robots before assembly. The single-carrier rule removes a redundant pickup assignment when it can identify the robot routed onward to the drop-off. The shared-drop-off rule spreads coincident destinations in eligible cases. The task evaluator monitors the original zones during execution, including when a guard changes a waypoint.

Execution feedback

The operator-chat implementation distinguishes three mechanisms: remediation after an execution error, repair informed by a post-execution state check, and mission supervision that can request additional work within configured limits. State verification reads robot poses and formation status. The results report which mechanisms were invoked in the selected runs.

5 Level 2: Fleet Coordination

MAPF backends

The PBS and LNS2 implementations accept robot identifiers and a goal for each robot through the same action interface. They publish scheduled waypoints to the robots. The simulation runs reported here use LNS2, which checks static reachability with its footprint model before planning.

Supporting components

A formation specification names a leader, followers and per-follower offsets in the leader's frame. The coordination layer excludes active followers from independent MAPF assignment and represents the group through the leader's enlarged planning footprint. This approximation is distinct from planning with the full, orientation-dependent group shape. BehaviorTree.CPP [23] runs the mission tree.

Operational meaning of task completion

The evaluator samples robot positions during execution at 5 Hz. A visit task completes when a pose sample shows a robot within 1.5 m of the target. For a carry task, the evaluator records the first robot it detects in the pickup zone and marks the task complete when it later detects that same robot in the drop-off zone. Each transition requires a single in-zone sample. The evaluator records zone presence independently of component status reports. Physical load handling is outside the metric.

6 Level 3: Per-Robot Execution

Each robot runs a namespaced Robot Operating System 2 (ROS 2) Humble [24] Nav2 [25] stack with staggered startup. Stage supplies the poses used for navigation and task evaluation. Adaptive Monte Carlo localization (AMCL) is disabled.

7 Experimental Design

The analysis uses the saved benchmark outputs, plans, logs and derived metrics at repository revision `cab6312`.

E1 — offline plan validation

The benchmark has 44 cases in twelve categories covering navigation, parallel and sequential composition, formation, disbanding, idle, escalation and edge cases. Each reported model and decoding condition contains five scored attempts per case at temperature 0.1. The parser extracts the plan and applies structural checks. Case validators then test selected requirements such as robot identifiers, proximity to a specified destination, and step ordering. A pass means that the response satisfies those checks. No robots are run in E1.

Schema-off requests use unconstrained text generation and parse a JSON plan from the result. Schema-on requests pass the plan schema as the response format. Neither offline condition submits tool definitions. The local sweep used Ollama, a

16 GB RTX 5060 Ti, a 32,768-token context, and a 2,048-token output limit. The local manifest records partial CPU offload for Mistral Small 3.2. Hosted runs used OpenRouter with an 8,192-token output limit. These are comparisons of the logged configurations, which differ in model, serving stack, and sometimes provider or token budget.

Hosted transport errors were handled by rerunning all five attempts of the affected case and replacing that case's original records. The raw records log replacements for 18 schema-off and 44 schema-on Nemotron cases, nine cases in each 397B condition, and 17 schema-on hosted 9B cases. These cells can mix provider routes. Thus 220 is the number of scored records per cell. The harness, its prompt builders, per-case prompts, responses, and scoring scripts support reanalysis of the saved outputs. Exact model snapshots and complete per-request serving configurations are unavailable.

E2 — task-evaluator completion in simulation

M2 consists of four visit tasks and one carry task. M3 consists of six visit tasks and three carry tasks, two of which share a drop-off location (Fig. 3). The selected descriptive series contains five scripted M2 runs with fixed assignments, five M2 runs using hosted Qwen3.5-9B, five M2 runs using hosted Qwen3.5-397B-A17B (cell B), and three M3 runs using the 397B model (cell B-M3). The scripted driver uses the same coordination and execution layers. Model, routing, allocation and software version vary across the reported cells.

Selection depended on outcome in B/M2. The development record reports that three original runs completed all five tasks, while runs `20260818_085836` and `20260818_090145` each stalled at 4/5. After the single-carrier guard was introduced, those two runs were replaced by `20260818_093216` and `20260818_093704`. The selected B/M2 cell thus combines three original successes and two successful repeats. The bags of the two superseded runs are absent from the public archive, so their scores are taken from the run history. The 18-run analysis estimates neither a campaign-wide completion rate nor a guard effect.

The archived session files and logs identify the included runs. A recorded base commit does not identify the uncommitted changes present in a run. The comparison uses the task evaluator recorded in `/tasks/state` and the target task list defined for each mission. For completed runs, the task-completion interval runs from the first recorded task-state sample to the first sample in which the last target is marked complete, using simulation time. Language-loop duration uses the final cumulative wall-clock counter in each language run's mission journal.

Table 2 Provenance and analysis populations for the simulation evidence

Series	Runs	Inclusion and evidence
Selected M2/M3	18	Four cells. Session IDs and base commits are recorded. All 18 worktrees were dirty. Thirteen runs have a language journal, of which eight have per-phase checkpoints.
Superseded B/M2	2	Two runs at 4/5 tasks replaced after a software change. The three original successes remain in the selected cell.
M3 before/after	2 additional	The pre-guard runs have logs and derived metrics but no decision journal. The three post-guard runs are already counted in the B-M3 cell above.
M4 formation	4 recorded attempts	Two retained qualitative cases (`20260818_140918`, `20260819_073009`). A pilot (`20260818_135527`) stopped early and a 42 s recording (`20260819_072231`) with no motion was discarded. M4 is outside the task-completion scoring for M2/M3.

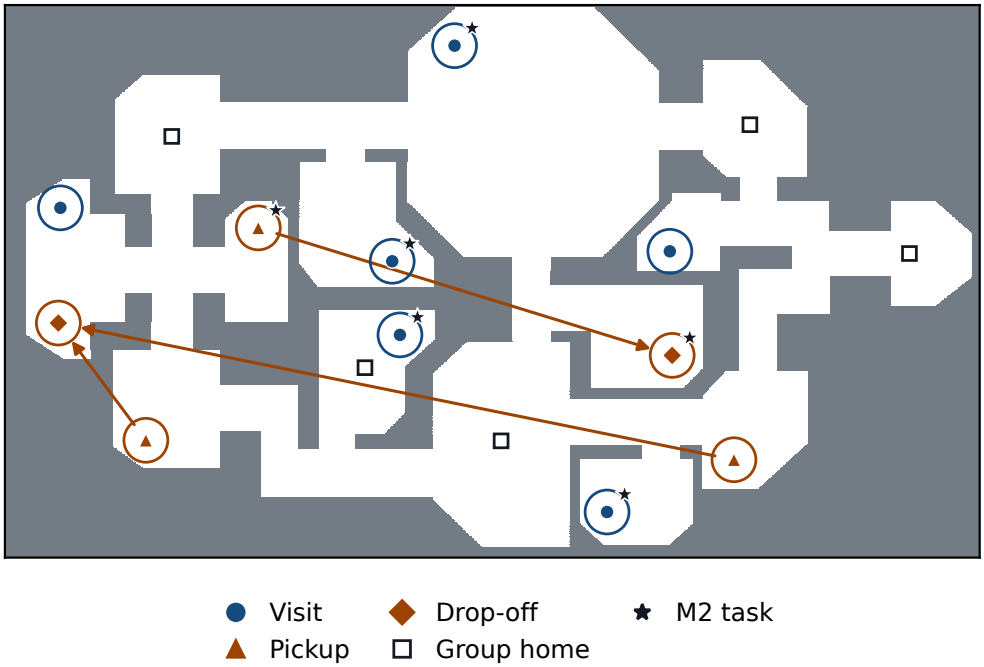


Fig. 3 The `amongus` map (66.6 × 37.75 m, 0.05 m cells) and the eleven distinct zones of radius 1.5 m used by the nine M3 tasks. Circles show the zone boundaries and center markers identify visit, pickup, and drop-off zones. Arrows link pickup and drop-off locations. Stars mark the M2 subset. Squares mark the home locations of the five groups of four robots

8 Results

8.1 Assignment defects and shared-destination conflicts

Duplicate-carrier assignments

Three of the 13 selected language runs have a saved plan that sends an extra robot to the pickup. In two further runs, the single-carrier guard counter records a correction and the saved plan already reflects it. All three model-9b runs with a visible extra assignment completed M2. Both B/M2 replacements also completed M2 after the guard removed an extra pickup assignment. These are observed interventions and outcomes in the selected series.

Shared-drop-off execution conflict

M3 assigns two carry tasks to the same drop-off center (Fig. 3), so both carriers can be directed to the same point. Both archived pre-guard runs (`20260815_153035` and `20260815_154541`) complete 8/9 tasks. Each of the three B-M3 runs completes 9/9. In run `20260818_111548` the guard changed the common target $(-29.65, -2.85)$ m to $(-28.90, -2.85)$ m for robot 3 and $(-30.40, -2.85)$ m for robot 16. Both goals are 0.75 m from the specified center and inside the original evaluator zone. The logs show spreading in all three post-guard runs. The two series differ in software version and provider routing, and the pre-guard journals are absent. These records document the intervention and outcomes without isolating its effect.

8.2 Case study: an assembled formation with a rejected route

In M4 run `20260819_073009`, the operator asked for a column. The model returned `offsets_x=[0,0,0]` and `offsets_y=[1.5,-1.5,3.0]`, a line abreast with a nominal span of 4.5 m between the outer centers. The log records activation with those offsets. On the following move, LNS2 reported no static path for the leader with the footprint approximation on the planning grid used for that request. The derived status metrics report 6.5 s from the first active sample to the first stable sample and no broken samples while active.

The plan passed structural validation and activated a formation whose shape differed from the request. The route was then rejected under the planner's approximation. In M4 run `20260818_140918`, the recorded plan used column offsets `offsets_x=[-1.5,-3.0,-4.5]`. These two records document different outcomes of plan generation.

8.3 Local success reports and the original request

In M4 run `20260818_140918`, MAPF reported *All agents arrived* while robot 0, part of the original request, remained at its starting position. The development record traces the mismatch to a replan that removed robot 0 from the set used to check arrival.

In a later step of the same run, the planner substituted a free cell for a blocked goal. The formation manager reported robot 3 at 1.97 m from its required slot, outside the 0.50 m tolerance. The executor learned of the refusal at its step timeout. These are two observations from one trace.

8.4 Offline plan validation results

Table 3 reports a response-level pass rate and the share of cases with at least one pass in five attempts. Table 4 breaks down the 259 failed responses among the 1,100 local schema-off responses. For Mistral Small 3.2, the same offsets-arity message accounts for 20 of 47 failures.

Table 3 Offline validation on 44 cases with five scored responses per case and condition. Call rate is the percentage of 220 responses that passed. Case rate is the percentage of 44 cases with at least one pass in five attempts. Model sizes are nominal; for mixture-of-experts models they are total/active parameters. n/d denotes an undisclosed size. A dash denotes an unreported condition

		Schema off		Schema on	
Model	Size	Call (%)	Case (%)	Call (%)	Case (%)
Local					
Qwen3.5-4B	4B	70.0	93.2	77.7	95.5
Gemma 3 12B	12B	71.4	72.7	—	—
Qwen2.5-14B	14B	75.0	81.8	75.5	81.8
Qwen3.5-9B	9B	87.3	95.5	19.1	22.7
Mistral Small 3.2	24B	78.6	81.8	—	—
Hosted					
Qwen3.5-9B	9B	90.5	97.7	51.4	61.4
Gemini 2.5 Flash	n/d	89.5	90.9	80.9	86.4
Nemotron 3 Ultra	550B/55B	90.5	97.7	88.6	95.5
Qwen3.5-397B-A17B	397B/17B	93.2	100.0	95.0	100.0

Table 4 Failure categories for local schema-off responses in E1. The scoring script assigns one category per failed response using a fixed priority order over error messages

Category	Failed responses
Formation fields or staging	82
Plan structure	81
Goal checks	34
Unparseable JSON	34
Other validator messages	22
Escalation reason	5
Robot identifiers	1
Total	259

8.4.1 Decoding configurations

When a response schema was requested, the local 9B score was 19.1%, versus 87.3% without one. The local 4B score was 77.7%, versus 70.0%, and the local 14B condition gained one passing response out of 220. Hosted differences also vary in direction. The 397B/17B scores were 95.0% with a schema and 93.2% without one, while Gemini scored 80.9% and 89.5%, respectively. These comparisons describe the logged configurations.

8.5 Task-evaluator completion in the selected series

All 18 included runs satisfy their target task conditions (Table 5). The B/M2 row combines the three successful original runs with the two successful replacements.

Table 5 Selected M2/M3 runs. Task counts are pose-based outcomes per run. The task interval uses simulation seconds and is reported as mean and sample standard deviation (SD). Language-loop time is the mean wall-clock duration, including auxiliary calls and retries. The version log reports that 3/5 original B/M2 runs completed all tasks. The two runs that stopped at 4/5 tasks were replaced after a software change

Cell	Mission	Full runs	Tasks per run	Task interval (sim s) mean ± SD	Language loop (wall s)
no-LLM	M2	5/5	5/5	203.5 ± 10.6	0
model-9b	M2	5/5	5/5	252.2 ± 26.6	9.5
B	M2	5/5	5/5	260.0 ± 16.5	31.6
B-M3	M3	3/3	9/9	587.7 ± 67.0	51.2

Timing and fleet activity

The M2 mean task interval ranges from 203.5 to 260.0 simulation seconds. The mean language-loop duration ranges from 0 to 31.6 wall-clock seconds. These intervals start at different events and use different clocks. Recorded odometry summaries show that 9–10 distinct robots per selected run traveled at least 0.5 m during the recording.

Observed mechanism use

None of the 13 selected language runs invoked remediation or verification repair, and none triggered the active-formation guard or chat-level occupancy rewrites. Each logged one supervision step. Twelve runs made no auxiliary calls. One B-M3 run (`20260818_113030`) called `list_tasks` once and `verify_plan_execution_state` twice. The two additional M4 records contain 10 and 33 auxiliary calls.

9 Discussion

For RQ1, the same model label can have very different observed pass rates across generation settings. The benchmark evaluates saved plans against case criteria. It has no paired mission outcomes.

For RQ2, the observed corrections address different problems. The single-carrier rule changes the assignment. The shared-drop-off rule changes the two waypoints while retaining the original evaluator zone. The M4 formation trace adds a separate distinction between accepting formation offsets and finding a route for the resulting footprint.

For RQ3, the arrival report in M4 referred to the set remaining after replanning. Checking it against the original request exposed the missing robot. A success report should therefore identify the robots and goals it covers, so it can be compared with the original command.

10 Limitations

The simulation evidence covers one Stage map, two main missions and selected runs. The version log documents the B/M2 replacements. The M3 before/after series differ in software version and provider routing, so the comparison cannot isolate the guard's effect. The recorded commits do not capture uncommitted changes in the historical runs.

E1 repeats 44 case designs and provides a comparison of the logged model and serving configurations. Exact model revisions and complete serving settings are unavailable. The simulation task metric records zone presence. It does not measure physical cargo transfer, collision safety or peak concurrency. The M4 observations illustrate individual failures and do not estimate their frequency.

11 Conclusion

The recorded cases separate plan validity, task assignment, route planning and completion. An accepted plan in these records could still include an extra pickup assignment, produce the wrong formation shape, or lead to an arrival report for a reduced robot set. To audit execution, record the original request, the action actually sent, the robot and goal set covered by the status report, and the observed outcome.

Statements and Declarations

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

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Ethics Approval

Not applicable. This study did not involve human participants or animals.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Data Availability

The benchmark outputs, session metadata, logs, derived metrics, and analysis scripts are in the results directory at the archived repository revision. The recorded bags are available in the public results archive, `llm_swarm_results_2026-11-04.tar.zst` (135 MB). Its SHA-256 checksum is `3866f1fce6254243986b0052e2fc0d6039eadc351e10c460f4ed0feec69f740d`. A per-file manifest accompanies the archive. With the archive unpacked, run the repository's `paper/results/scripts/analyze_sessions.py` with `--sessions` pointing to the unpacked `results/e2_mission_series` and `--json` to export per-run metrics. The two superseded B/M2 bags are absent. Their scores and replacement identifiers are documented in the run history.

Code Availability

The ROS 2 stack, benchmark harness, and analysis scripts are available at repository revision `cab6312`. This revision records the analysis and documentation. It does not capture the uncommitted changes in the historical simulation runs. Licence declarations, where present, are in the individual packages.

Author Contributions

A.T.: Conceptualization, Methodology, Software, Investigation, Data curation, Writing—original draft, Writing—review and editing, Project administration. D.V.: Software, Investigation, Writing—review and editing. E.S.: Conceptualization, Methodology, Software, Investigation, Supervision, Project administration, Writing—review and editing. S.B.: Software, Investigation, Visualization. A.Mal.: Software, Validation. A.Malol.: Supervision, Conceptualization, Writing—review and editing.

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